

Statistical Modelling Coursework 2

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Question 1:

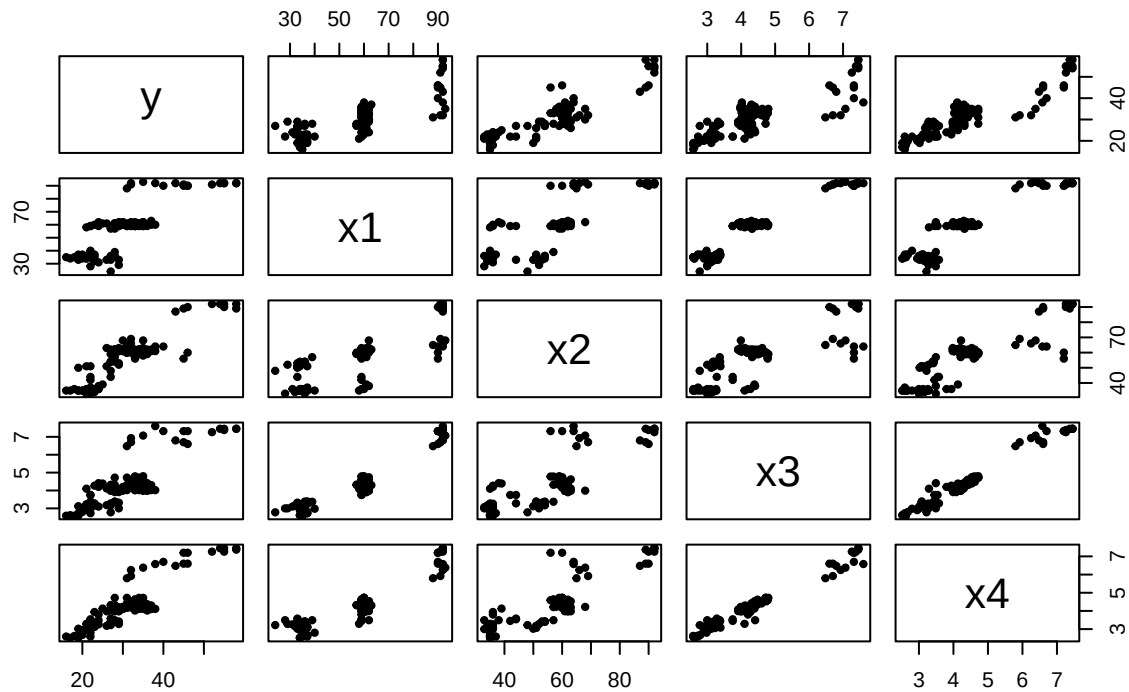
i).

```
summary(petrol) # Obtain summary of data
```

```
##           y           x1           x2           x3           x4
##  Min.    :16.00   Min.    :24.00   Min.    :33.0   Min.    :2.590   Min.    :2.520
##  1st Qu.:25.00   1st Qu.:37.00   1st Qu.:50.0   1st Qu.:3.370   1st Qu.:3.430
##  Median :30.00   Median :60.00   Median :60.0   Median :4.110   Median :4.110
##  Mean   :30.58   Mean   :57.32   Mean   :56.5   Mean   :4.307   Mean   :4.253
##  3rd Qu.:34.00   3rd Qu.:61.00   3rd Qu.:62.0   3rd Qu.:4.480   3rd Qu.:4.450
##  Max.    :58.00   Max.    :93.00   Max.    :92.0   Max.    :7.600   Max.    :7.450
```

```
pairs(petrol,
      main = "Scatterplot matrix for petrol data",
      pch = 20) # Plotting the Scatterplot matrix
```

Scatterplot matrix for petrol data



```
round(cor(petrol), 4)
```

```
##           y      x1      x2      x3      x4
## y  1.0000 0.8078 0.8778 0.8505 0.9072
## x1 0.8078 1.0000 0.8065 0.9422 0.9265
## x2 0.8778 0.8065 1.0000 0.7698 0.8310
## x3 0.8505 0.9422 0.7698 1.0000 0.9838
## x4 0.9072 0.9265 0.8310 0.9838 1.0000
```

The scatterplot matrix appears to show strong positive relationships between y and all four explanatory variables. The strongest relationship is between y and x_4 , with a correlation of 0.9072. The relationships between y and x_2 , x_3 , and x_1 are also strong, with correlations of 0.8778, 0.8505, and 0.8078, respectively. However, the plots show slight upward curvature, suggesting that the relationships may not be perfectly linear on the original scale. This indicates that it may be useful to consider using $\log(y)$ in the model to see whether the relationships become more linear. Therefore, all four explanatory variables could be useful for modelling either y or $\log(y)$, depending on which response scale is more suitable.

There also appears to be some clustering in the plots, but this seems to be due to some explanatory variable values falling into groups, rather than indicating a problem with the data.

The explanatory variables are also strongly related to each other. In particular, x_3 and x_4 have a very strong positive linear relationship, with a correlation of 0.9838. This could cause multicollinearity when fitting a model; making coefficient estimates unstable since the model would struggle to distinguish between the individual effects of x_3 and x_4 . Most other pairs of explanatory variables also show positive correlations that are greater than 0.8000, although x_2 and x_3 have the weakest relationship, with a correlation of 0.7698.

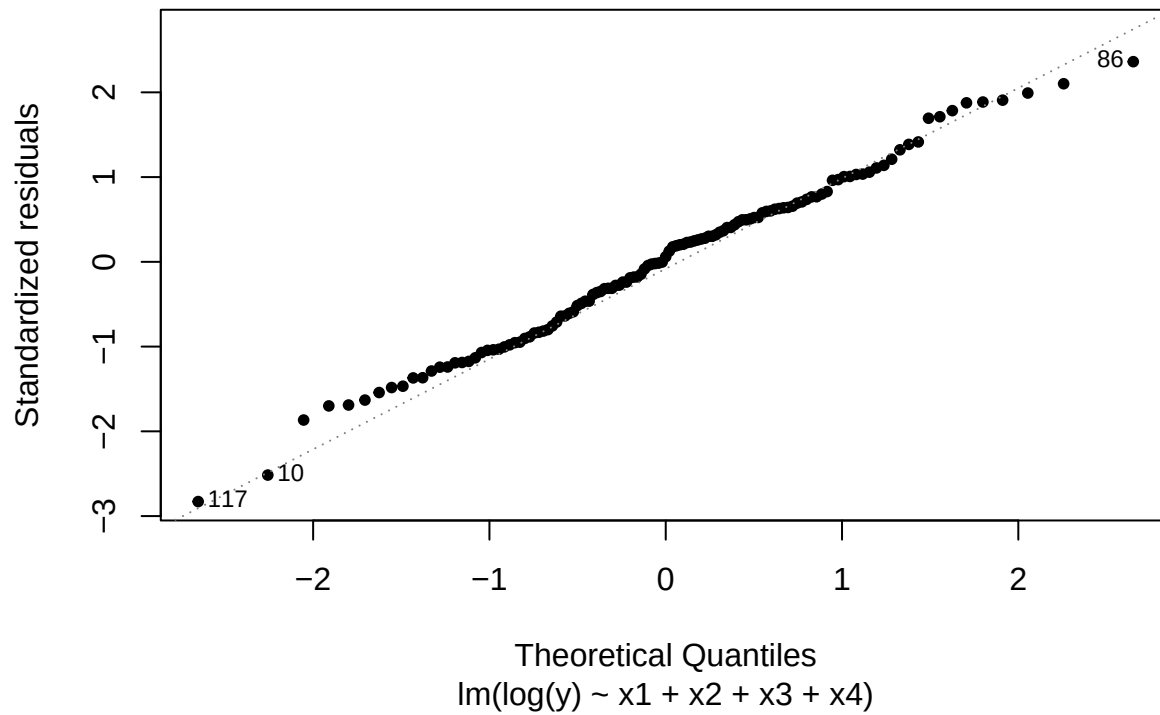
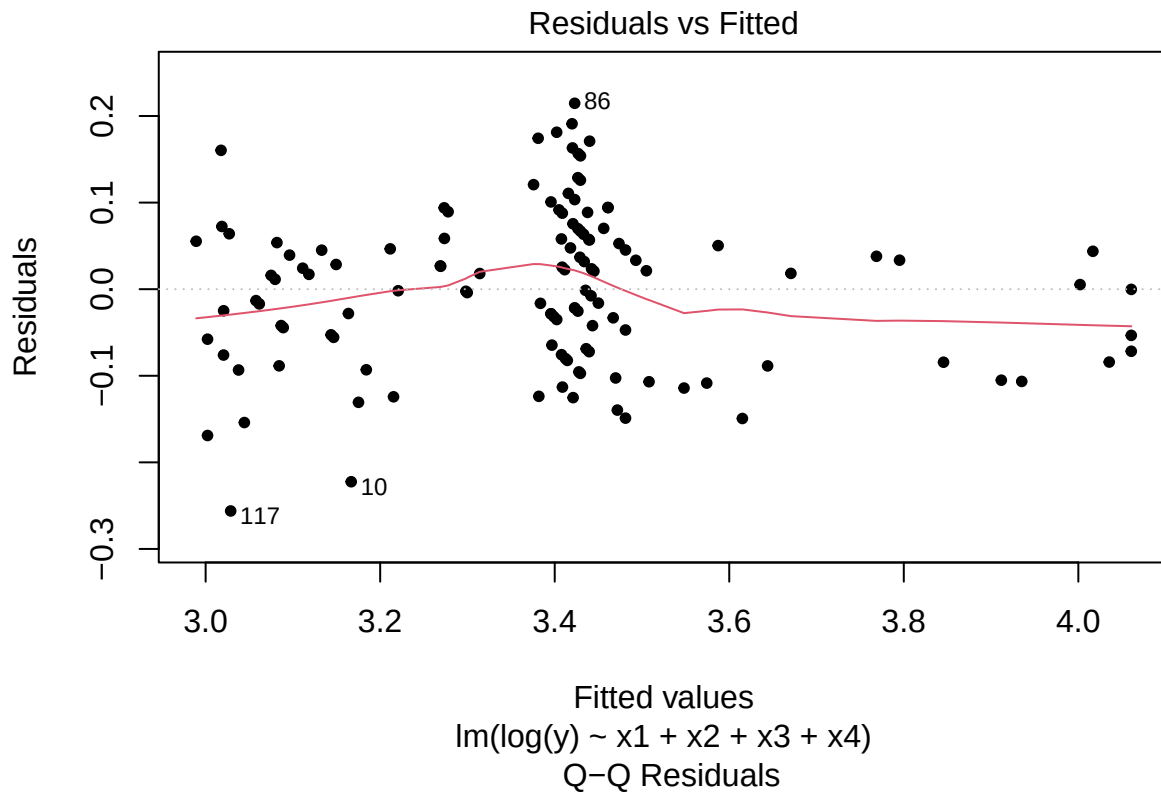
By visual inspection, there do not appear to be any obvious outliers, although the clustering in the data may make them harder to identify by eye alone.

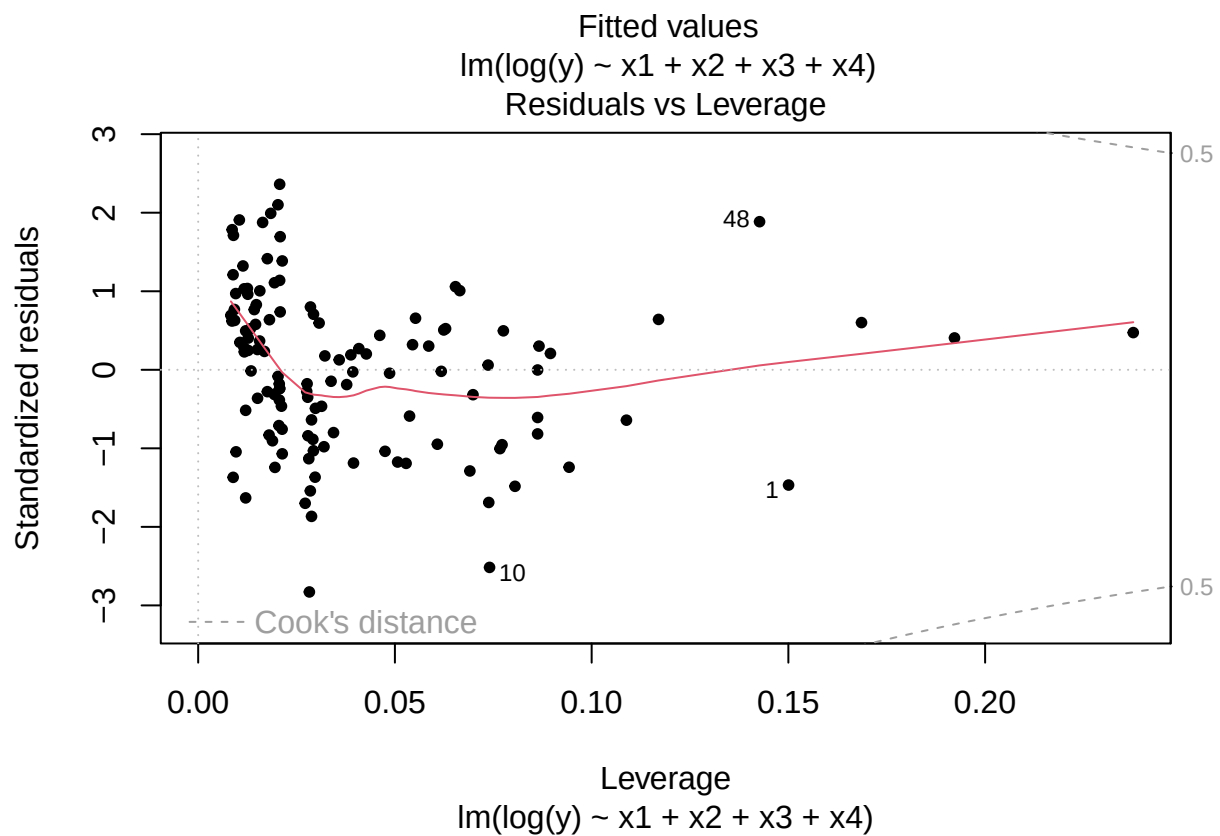
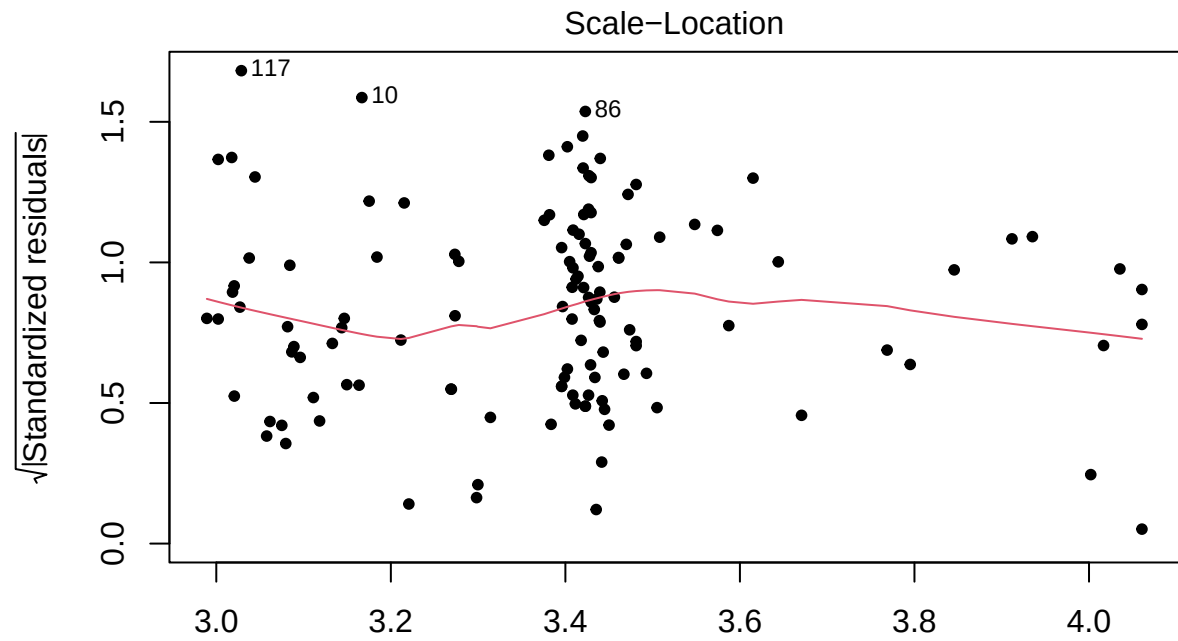
ii).

Postulated multiple linear regression model:

$$\log(y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \epsilon_i, \quad \epsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2).$$

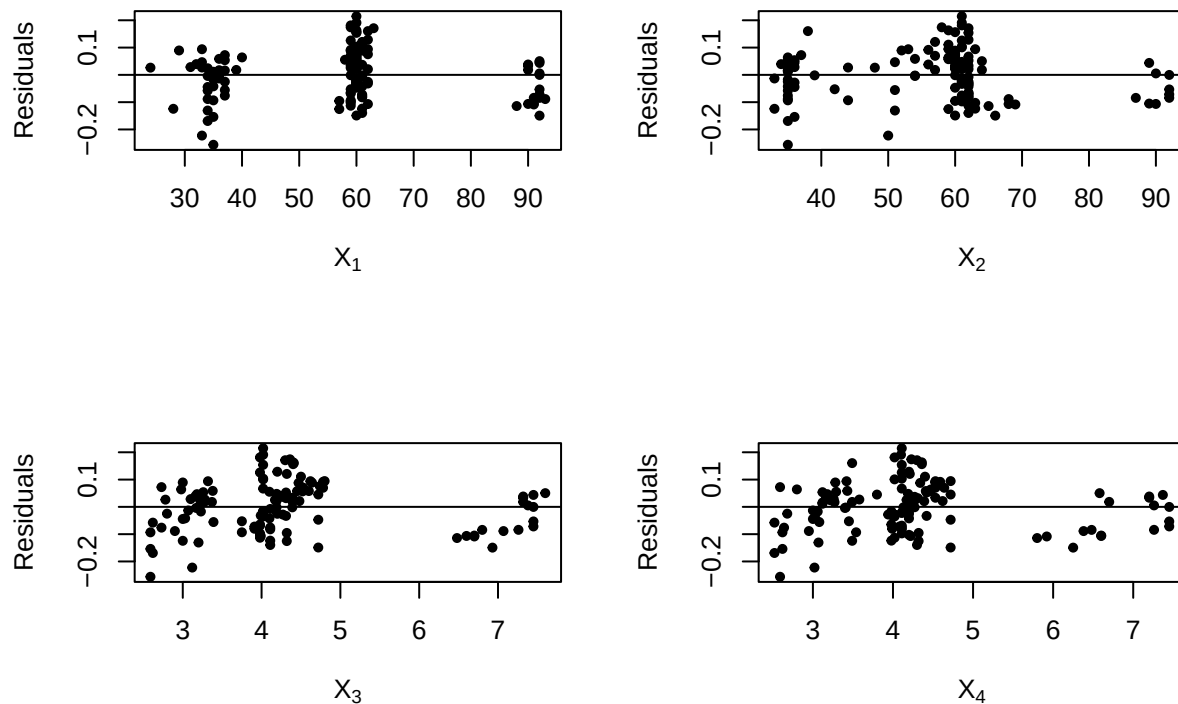
```
original.lm <- lm(log(y) ~ x1 + x2 + x3 + x4, data = petrol) # Creating the MLRM
plot(original.lm,
      pch=20)
```





```
# Plotting residual vs _1, x2, ...
par(mfrow = c(2, 2))
plot(petrol$x1, original.lm$resid, xlab = expression(X[1]), ylab = "Residuals", pch = 20)
abline(h = 0)
plot(petrol$x2, original.lm$resid, xlab = expression(X[2]), ylab = "Residuals", pch = 20)
abline(h = 0)
```

```
plot(petrol$x3, original.lm$resid, xlab = expression(X[3]), ylab = "Residuals", pch = 20)
abline(h = 0)
plot(petrol$x4, original.lm$resid, xlab = expression(X[4]), ylab = "Residuals", pch = 20)
abline(h = 0)
```



```
par(mfrow = c(1, 1))

# VIF check for the model: log(y) ~ x1 + x2 + x3 + x4

vif_x1 <- 1 / (1 - summary(lm(x1 ~ x2 + x3 + x4, data = petrol))$r.squared)

vif_x2 <- 1 / (1 - summary(lm(x2 ~ x1 + x3 + x4, data = petrol))$r.squared)

vif_x3 <- 1 / (1 - summary(lm(x3 ~ x1 + x2 + x4, data = petrol))$r.squared)

vif_x4 <- 1 / (1 - summary(lm(x4 ~ x1 + x2 + x3, data = petrol))$r.squared)

round(vif_x1, 4)

## [1] 11.9273
round(vif_x2, 4)

## [1] 5.6157
round(vif_x3, 4)

## [1] 66.1382
round(vif_x4, 4)

## [1] 60.9387
```

```
summary(original.lm) # Obtain summary of the fit of the MLRM

##
## Call:
## lm(formula = log(y) ~ x1 + x2 + x3 + x4, data = petrol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.256122 -0.071691  0.005317  0.057491  0.214689
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.4468643  0.0344701  70.985  < 2e-16 ***
## x1           0.0002837  0.0016195   0.175  0.86124
## x2           0.0066139  0.0013703   4.826 4.13e-06 ***
## x3          -0.1704730  0.0528239  -3.227  0.00161 **
## x4           0.3019134  0.0542632   5.564 1.63e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09185 on 120 degrees of freedom
## Multiple R-squared:  0.8772, Adjusted R-squared:  0.8731
## F-statistic: 214.2 on 4 and 120 DF,  p-value: < 2.2e-16

round(summary(original.lm)$coefficients, 4)

##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.4469      0.0345 70.9851  0.0000
## x1              0.0003      0.0016  0.1752  0.8612
## x2              0.0066      0.0014  4.8265  0.0000
## x3             -0.1705      0.0528 -3.2272  0.0016
## x4              0.3019      0.0543  5.5639  0.0000
```

Therefore, our fitted model is:

$$\widehat{\log(y_i)} = 2.4469 + 0.0003x_{i1} + 0.0066x_{i2} - 0.1705x_{i3} + 0.3019x_{i4}.$$

The full model has $R^2 = 0.8772$, so it explains a large proportion of the variation in $\log(y)$. From the coefficient table, x_2 , x_3 and x_4 appear to be statistically significant in the model. In contrast, x_1 does not appear to be statistically significant at the 5% level once the other explanatory variables are included, as shown by its large p-value of 0.8612.

The residuals versus fitted values plot shows a roughly constant vertical spread of residuals about the line $y = 0$, with no clear curvature or fanning pattern. Therefore, there is no strong visual evidence to suggest heteroscedasticity. The residuals do appear clustered, but this is expected because the data values occur in specific groups rather than being evenly spread. Hence, the clustering does not by itself indicate a systematic residual pattern.

Since this is a multiple linear regression model, it is also useful to check the residuals against each individual explanatory variable, to look for any hidden problems that may not be visible in the residuals versus fitted values plot. The residual plots against $x_1, \dots, x_4$ again show clustering, but within each cluster the residuals appear to be scattered randomly around zero, with no strong systematic pattern evident.

The Scale-Location plot shows the square root of the standardised residuals against the fitted values. The red line is fairly flat, with no clear fan shape, so there is no strong evidence of non-constant variance. Although there is some minor variation in spread across the fitted values, the overall pattern does not suggest severe heteroscedasticity.

The residuals versus leverage plot is used to identify observations that may have high influence on the fitted model. Most observations have low leverage and lie close to the main cluster. Observations such as 1, 10 and 48 have relatively higher leverage or larger residuals, but they do not appear to exceed a Cook's distance of 0.5. Therefore, there is no strong evidence of any highly influential observation in the fitted model.

The Q-Q plot follows an approximately straight line, with only slight deviations in the tails. This suggests that the normality assumption for the error term is reasonably well satisfied.

From the correlation matrix in part (i), x_3 and x_4 were highly correlated, suggesting that variance inflation factors should be calculated. The large VIF values of 66.1382 for x_3 and 60.9387 for x_4 confirm that strong multicollinearity is present. Therefore, the coefficient estimates for x_3 and x_4 should be interpreted carefully, since their standard errors may be inflated and the estimates may be unstable.

Overall, the model appears to be a reasonable fit, due to the high R^2 value, the approximately linear Q-Q plot, and the lack of a strong systematic pattern in the residual plots. However, the large p-value for x_1 suggests that it is a likely candidate for removal during backward elimination.

iii).

Postulated MLRM:

$$\log(y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \epsilon_i, \quad \epsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2).$$

```
original_model1 <- lm(log(y) ~ x1 + x2 + x3 + x4, data = petrol) # Assigning lm
# Automatically perform backwards elimination
simp_model1 <- step(original_model1, direction = "backward")
```

```
## Start:  AIC=-592
## log(y) ~ x1 + x2 + x3 + x4
##
##      Df Sum of Sq    RSS    AIC
## - x1    1  0.000259 1.0126 -593.97
## <none>                1.0124 -592.00
## - x3    1  0.087865 1.1002 -583.60
## - x2    1  0.196526 1.2089 -571.82
## - x4    1  0.261167 1.2735 -565.31
##
## Step:  AIC=-593.97
## log(y) ~ x2 + x3 + x4
##
##      Df Sum of Sq    RSS    AIC
```

```
## <none>          1.0126 -593.97
## - x3      1    0.13426 1.1469 -580.41
## - x2      1    0.27272 1.2854 -566.16
## - x4      1    0.29247 1.3051 -564.25

# Results
summary(simp_model1)

##
## Call:
## lm(formula = log(y) ~ x2 + x3 + x4, data = petrol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.255579 -0.070106  0.003559  0.056617  0.216080
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.445847   0.033841  72.274 < 2e-16 ***
## x2           0.006735   0.001180   5.709 8.28e-08 ***
## x3          -0.164700   0.041120  -4.005 0.000107 ***
## x4           0.298526   0.050498   5.912 3.21e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09148 on 121 degrees of freedom
## Multiple R-squared:  0.8771, Adjusted R-squared:  0.8741
## F-statistic: 288 on 3 and 121 DF, p-value: < 2.2e-16

coef(simp_model1)

##      (Intercept)          x2          x3          x4
## 2.445847107  0.006734615 -0.164700485  0.298526417

formula(simp_model1)

## log(y) ~ x2 + x3 + x4
```

We used AIC to decide which variables to remove during backward elimination. From the calculations above, the AIC decreased from -592.00 to -593.97 after removing x_1 . This was the largest reduction in AIC when removing 1 variable, so x_1 was removed from the model. We then repeated the procedure, however removing any further variable increased the AIC, suggesting that no further simplification was beneficial for the model. Therefore, the only variable removed was x_1 , giving the final model for $\log(y)$ as:

$$\log(y_i) = \beta_0 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \epsilon_i, \quad \epsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2).$$

We can also see that removing x_1 increased the adjusted R^2 value from 0.8731 to 0.8741. Since we are comparing models, we put a higher emphasis on adjusted R^2 values as they give more information about the overall fit of the model (by taking into account the number of explanatory variables). Therefore, removing x_1 seems to have slightly improved the model.

```
# VIF calculation

vif_x2 <- 1 / (1 - summary(lm(x2 ~ x3 + x4, data = petrol))$r.squared)

vif_x3 <- 1 / (1 - summary(lm(x3 ~ x2 + x4, data = petrol))$r.squared)
```



```
vif_x4 <- 1 / (1 - summary(lm(x4 ~ x2 + x3, data = petrol))$r.squared)

round(c(x2 = vif_x2, x3 = vif_x3, x4 = vif_x4), 4)
```

```
##      x2      x3      x4
## 4.1958 40.4009 53.2020
```

It should be noted that both x_3 and x_4 remain in the selected model and their new VIFs, 40.4009 and 53.2020 respectively, still indicate strong multicollinearity between these variables causing their coefficient estimates to have high variance.

We also fitted this model to the data to give us our final fitted model for $\log(y)$:

$$\widehat{\log(y_i)} = 2.4458 + 0.0067x_{i2} - 0.1647x_{i3} + 0.2985x_{i4}.$$

iv).

If we add x_3^2 and x_4^2 as variables to the model found in (iii), our postulated MLRM is:

$$\log(y_i) = \beta_0 + \beta_2x_{i2} + \beta_3x_{i3} + \beta_4x_{i4} + \beta_5x_{i3}^2 + \beta_6x_{i4}^2 + \epsilon_i, \quad \epsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2).$$

```
original_model2 <- lm(log(y) ~ x2 + x3 + x4 + I(x3^2) + I(x4^2),
                      data = petrol) # Assigning the model
```

```
# Automatically perform backwards elimination
simp_model2 <- step(original_model2, direction = "backward")
```

```
## Start:  AIC=-609.31
## log(y) ~ x2 + x3 + x4 + I(x3^2) + I(x4^2)
##
##           Df Sum of Sq    RSS    AIC
## - I(x4^2)  1  0.002589 0.87009 -610.93
## - x3       1  0.002701 0.87020 -610.92
## <none>                      0.86750 -609.31
## - I(x3^2)  1  0.036076 0.90358 -606.21
## - x4       1  0.036389 0.90389 -606.17
## - x2       1  0.103233 0.97073 -597.25
##
## Step:  AIC=-610.93
## log(y) ~ x2 + x3 + x4 + I(x3^2)
##
##           Df Sum of Sq    RSS    AIC
## - x3       1  0.00022 0.87031 -612.90
## <none>                      0.87009 -610.93
## - x2       1  0.10379 0.97387 -598.85
## - I(x3^2)  1  0.14255 1.01264 -593.97
## - x4       1  0.37523 1.24531 -568.12
##
## Step:  AIC=-612.9
## log(y) ~ x2 + x4 + I(x3^2)
##
##           Df Sum of Sq    RSS    AIC
## <none>                      0.87031 -612.90
## - x2       1  0.10369 0.97399 -600.83
## - I(x3^2)  1  0.27660 1.14690 -580.41
```

```
## - x4      1    0.50925 1.37955 -557.32

# Results
summary(simp_model2)

##
## Call:
## lm(formula = log(y) ~ x2 + x4 + I(x3^2), data = petrol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.193961 -0.061714  0.004261  0.048215  0.204880
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.018233   0.075216  26.832 < 2e-16 ***
## x2           0.004550   0.001198   3.797 0.000231 ***
## x4           0.351393   0.041761   8.414 9.25e-14 ***
## I(x3^2)      -0.019016   0.003066  -6.201 8.07e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08481 on 121 degrees of freedom
## Multiple R-squared:  0.8944, Adjusted R-squared:  0.8918
## F-statistic: 341.6 on 3 and 121 DF,  p-value: < 2.2e-16

coef(simp_model2)

##      (Intercept)          x2          x4      I(x3^2)
## 2.018233061  0.004550086  0.351392916 -0.019016060

formula(simp_model2)

## log(y) ~ x2 + x4 + I(x3^2)

# VIF calculation

vif_x2 <- 1 / (1 - summary(lm(x2 ~ x4 + I(x3^2), data = petrol))$r.squared)

vif_x4 <- 1 / (1 - summary(lm(x4 ~ x2 + I(x3^2), data = petrol))$r.squared)

vif_x3sq <- 1 / (1 - summary(lm(I(x3^2) ~ x2 + x4, data = petrol))$r.squared)

round(c(x2 = vif_x2, x4 = vif_x4, x3_squared = vif_x3sq), 4)

##           x2           x4 x3_squared
##    5.0376    42.3353    27.9436

# Plotting residual diagnostics

model_iv.lm <- lm(log(y) ~ x2 + x4 + I(x3^2), data = petrol)

par(mfrow = c(1, 2))

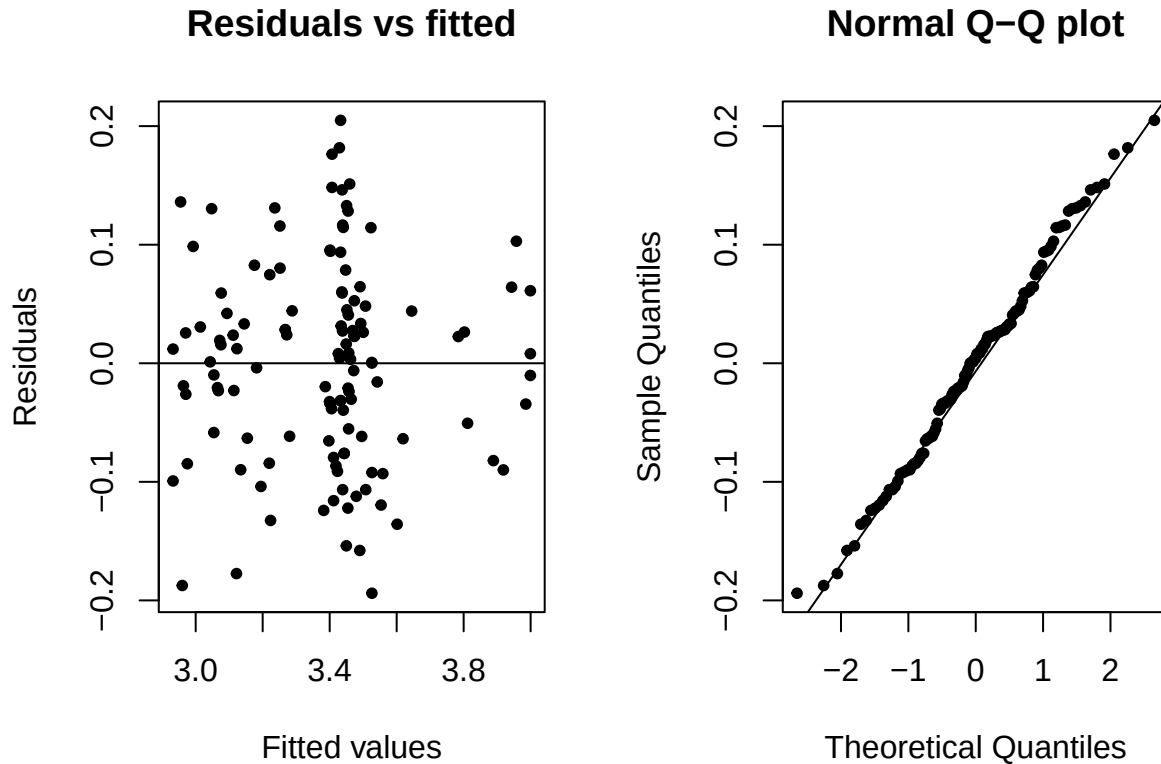
plot(model_iv.lm$fitted, model_iv.lm$resid,
      xlab = "Fitted values",
      ylab = "Residuals",
```

```

    main = "Residuals vs fitted",
    pch = 20)
abline(h = 0)

qqnorm(model_iv.lm$resid,
    main = "Normal Q-Q plot",
    pch = 20)
qqline(model_iv.lm$resid)

```



```
par(mfrow = c(1, 1))
```

Since backwards elimination was based on AIC results we chose the model with the lowest AIC value (AIC = -612.90). From the results of AIC we can see that backwards elimination removed x_4^2 first and then x_3 . This leaves us a final model of the form:

$$\log(y_i) = \beta_0 + \beta_2 x_{i2} + \beta_4 x_{i4} + \beta_5 x_{i3}^2 + \epsilon_i, \quad \epsilon_i \stackrel{ind}{\sim} N(0, \sigma^2).$$

Since x_3 and x_4 were highly correlated in the previous model, it is important to check whether the new model containing x_4 and x_3^2 also suffers from multicollinearity so we recalculated the VIFs. The VIF values were 42.3353 for x_4 and 27.9436 for x_3^2 , showing that this model still contains high levels of multicollinearity, although not as severe as in the model containing both x_3 and x_4 . Hence, the coefficients of x_4 and x_3^2 should still be interpreted carefully.

To assess the fit of the model, residual and Q-Q plots were produced. The residual plot shows no clear evidence of heteroscedasticity, so there is no strong visual evidence against the constant variance assumption. The middle of the Q-Q plot appears approximately linear, with only mild deviations in the tails, suggesting that the normality assumption is reasonably well satisfied. Overall, these diagnostics suggest that the model is a good fit, with limited areas of concern.

We also calculated the final fitted coefficients for the chosen model, giving the fitted model:

$$\widehat{\log(y_i)} = 2.0182 + 0.0046x_{i2} + 0.3514x_{i4} - 0.0190x_{i3}^2.$$

v).

```
# Adjusting data set to data points 1 - 85
model_iii <- lm(log(y) ~ x2 + x3 + x4,
               data = petrol,
               subset = 1:85)

# Using model.iii to Predict values for data points 86 - 125
log_prediction_iii <- predict(model_iii, newdata = petrol[86:125, ])
prediction_iii <- exp(log_prediction_iii) # Transforming back to raw scale
rmse_iii <- sqrt(mean((petrol$y[86:125] - prediction_iii)^2)) # Finding RMSE

# Adjusting data set to data points 1 - 85
model_iv <- lm(log(y) ~ x2 + x4 + I(x3^2),
               data = petrol,
               subset = 1:85)

# Using model.iv to Predict values for data points 86 - 125
log_prediction_iv <- predict(model_iv, newdata = petrol[86:125, ])
prediction_iv <- exp(log_prediction_iv) # Transforming back to raw scale
rmse_iv <- sqrt(mean((petrol$y[86:125] - prediction_iv)^2)) # Finding RMSE

print(rmse_iii)

## [1] 2.696027

print(rmse_iv)

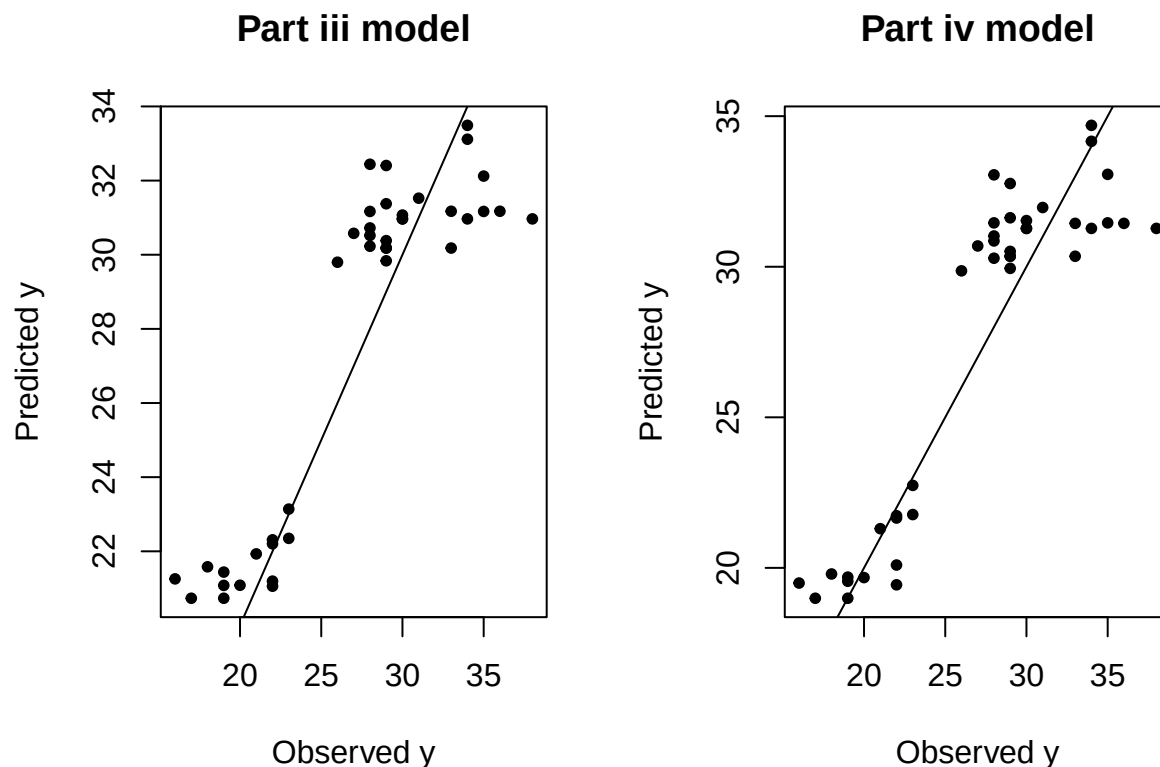
## [1] 2.525008

# Plotting residual diagnostics for (iii) and (iv)

par(mfrow = c(1, 2))

plot(petrol$y[86:125], prediction_iii,
     xlab = "Observed y",
     ylab = "Predicted y",
     main = "Part iii model",
     pch = 20)
abline(0, 1)

plot(petrol$y[86:125], prediction_iv,
     xlab = "Observed y",
     ylab = "Predicted y",
     main = "Part iv model",
     pch = 20)
abline(0, 1)
```



```
par(mfrow = c(1, 1))
```

When the models were trained using rows 1:85 and used to predict the held-out observations in rows 86:125, the model selected in part (iii) had an RMSPE of 2.6960 grams, compared with 2.5250 grams for the model selected in part (iv). Therefore, the model from part (iv) gives the smaller prediction error on this validation set.

However, this conclusion is based on a single training-validation split, so it does not prove that the model from part (iv) would always predict better. As an additional check, both models were also trained using rows 41:125 and used to predict rows 1:40. The RMSPE values were 2.7254 for model (iii) and 2.4806 for model (iv). This gives the same overall conclusion as the original validation split, increasing confidence that model (iv) has better predictive performance for this data set. However, since this is still based on a limited number of validation checks, the conclusion should be interpreted cautiously.

Predicted values were also plotted against observed values to check whether either model showed systematic prediction errors. For model (iii), the plot suggests that smaller values of y are often overpredicted, indicating some systematic prediction error. In contrast, the plot for model (iv) appears more balanced, with points lying more evenly around the line $y_{\text{predicted}} = y_{\text{observed}}$. This supports the RMSPE comparison and suggests that model (iv) gives better predictions on the raw scale.

Since both models show evidence of multicollinearity, this should not be the main deciding factor when choosing between them for prediction. Instead, the comparison should be based mainly on predictive performance on the held-out observations. Since model (iv) has the smaller RMSPE in both validation checks, it is preferred for prediction, while still noting that neither model is perfect.

The preferred model is:

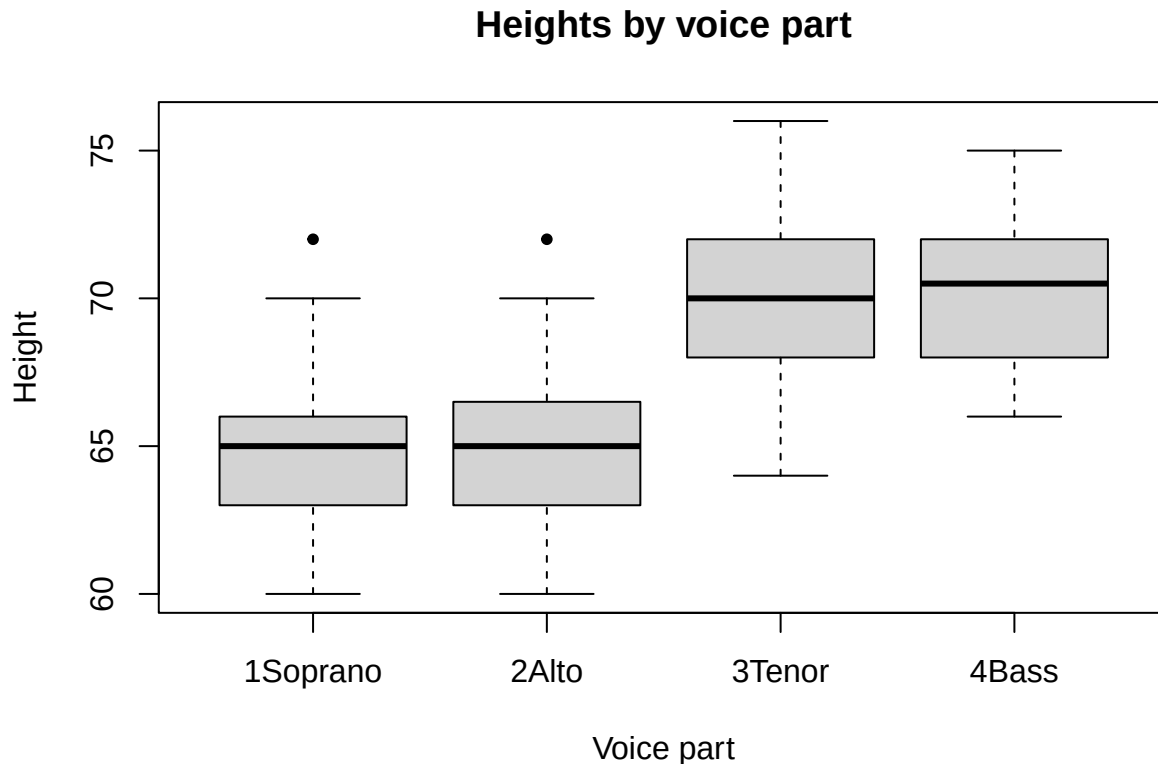
$$\widehat{\log(y_i)} = 2.0182 + 0.0046x_{i2} + 0.3514x_{i4} - 0.0190x_{i3}^2$$

Question 2)

i).

```
singers$voice <- factor(singers$voice) # Making voice a factor

boxplot(height ~ voice,
        data = singers,
        xlab = "Voice part",
        ylab = "Height",
        main = "Heights by voice part",
        pch = 20) # plotting box plots for all 4 voice parts
```



The boxplots suggest clear differences in height between two main groups of voice parts: Soprano/Alto and Tenor/Bass. Within each of these groups, however, the differences in median height are small. Soprano and Alto both have median heights of around 65 inches, while Tenor and Bass both have median heights of around 70 inches. This is consistent with the information given in the question, since Soprano and Alto are typically female voice parts, whereas Tenor and Bass are typically male voice parts. The whiskers for the four boxplots also appear to be of broadly similar lengths, suggesting that the spread of heights is reasonably similar across the groups.

There are possible high outliers in the Soprano and Alto groups, but these values do not appear extreme enough to suggest data errors or to change the overall pattern. The boxplots within each pair of voice parts are also fairly similar in shape, suggesting that most of the observed difference is between the Soprano/Alto and Tenor/Bass groups, rather than within those pairs. Although the Bass median is slightly higher than the Tenor median, the difference is small, so the boxplot alone does not provide strong evidence of a meaningful difference between these two groups.

ii).

```
anova_lm <- lm(height ~ voice, data = singers)
anova(anova_lm)
```

```
## Analysis of Variance Table
##
```

```
## Response: height
##           Df Sum Sq Mean Sq F value    Pr(>F)
## voice      3 940.85 313.617  43.214 < 2.2e-16 ***
## Residuals 126 914.42   7.257
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(anova_lm)
```

```
##
## Call:
## lm(formula = height ~ voice, data = singers)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.1714 -2.1714  0.2821  1.2821  7.2821
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  64.7179     0.4314 150.027 < 2e-16 ***
## voice2Alto    0.1987     0.6226   0.319    0.75
## voice3Tenor   5.4535     0.6272   8.694 1.58e-14 ***
## voice4Bass    5.6821     0.7409   7.669 4.09e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.694 on 126 degrees of freedom
## Multiple R-squared:  0.5071, Adjusted R-squared:  0.4954
## F-statistic: 43.21 on 3 and 126 DF,  p-value: < 2.2e-16
```

$$Y_{ij} = \mu_i + \epsilon_{ij}, \quad \epsilon_{ij} \stackrel{\text{ind}}{\sim} N(0, \sigma^2),$$

$$H_0 : \mu_{\text{Soprano}} = \mu_{\text{Alto}} = \mu_{\text{Tenor}} = \mu_{\text{Bass}},$$

$$H_1 : \text{at least one mean height is different.}$$

Our estimated regression coefficients are:

$$\hat{\beta}_0 = 64.7179$$

$$\hat{\beta}_{\text{Alto}} = 0.1987$$

$$\hat{\beta}_{\text{Tenor}} = 5.4535$$

$$\hat{\beta}_{\text{Bass}} = 5.6821.$$

Our calculated means for each group are:

$$\hat{\mu}_{\text{Soprano}} = 64.7179$$

$$\hat{\mu}_{\text{Alto}} = 64.7179 + 0.198 = 64.9166$$

$$\hat{\mu}_{\text{Tenor}} = 64.7179 + 5.4535 = 70.1714$$

$$\hat{\mu}_{\text{Bass}} = 64.7179 + 5.6821 = 70.4000$$

The ANOVA gives $F = 43.214$ with p-value $< 2.2 \times 10^{-16}$. Since this p-value is far below 0.05, we reject H_0 . Therefore, there is strong evidence that at least one mean height differs between the voice parts.

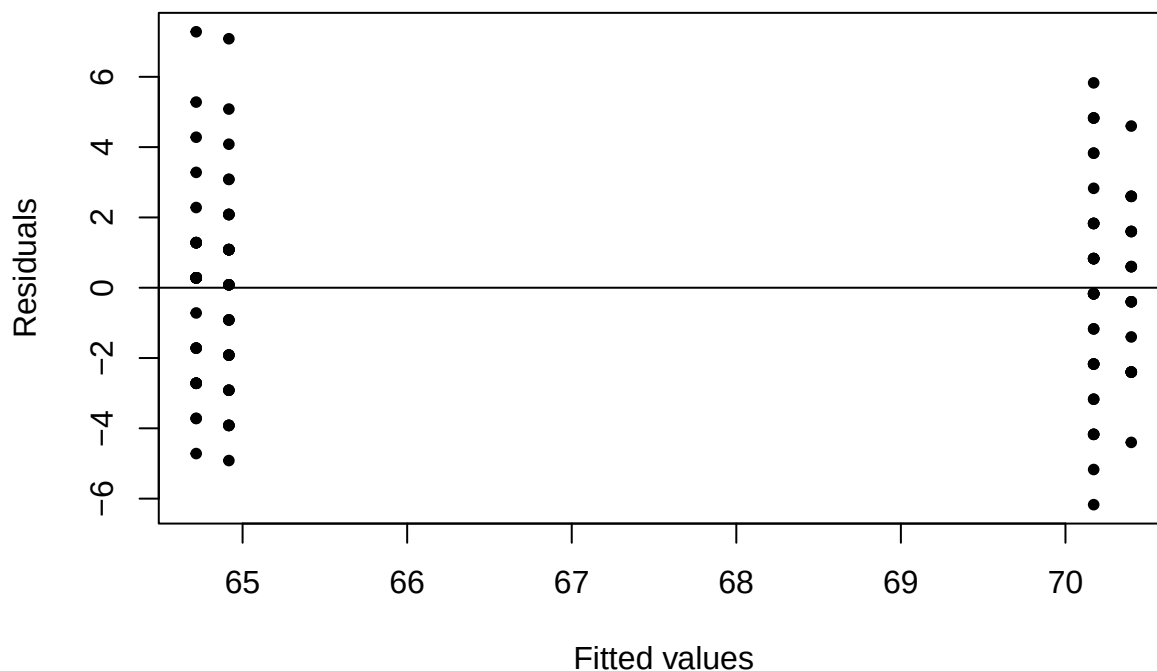
Using Soprano as the baseline group, the fitted mean height for Soprano is 64.72 inches. Alto is estimated to be 0.20 inches taller than Soprano, giving a fitted mean of 64.92 inches. However, this difference is not statistically significant, since the p-value for Alto is 0.75. Tenor and Bass are estimated to be 5.45 and 5.68 inches taller than Soprano, giving fitted means of 70.17 and 70.40 inches, respectively.

Although the ANOVA provides strong evidence that mean height differs between the voice parts, this does not provide sufficient evidence to conclude that female singers tend to over-report their height. Since the heights are self-reported, and no true measured heights are available for comparison it is hard to conclude that there is any evidence that female singers over report their height from the data provided.

iii).

```
# Plotting ANOVA diagnostic
```

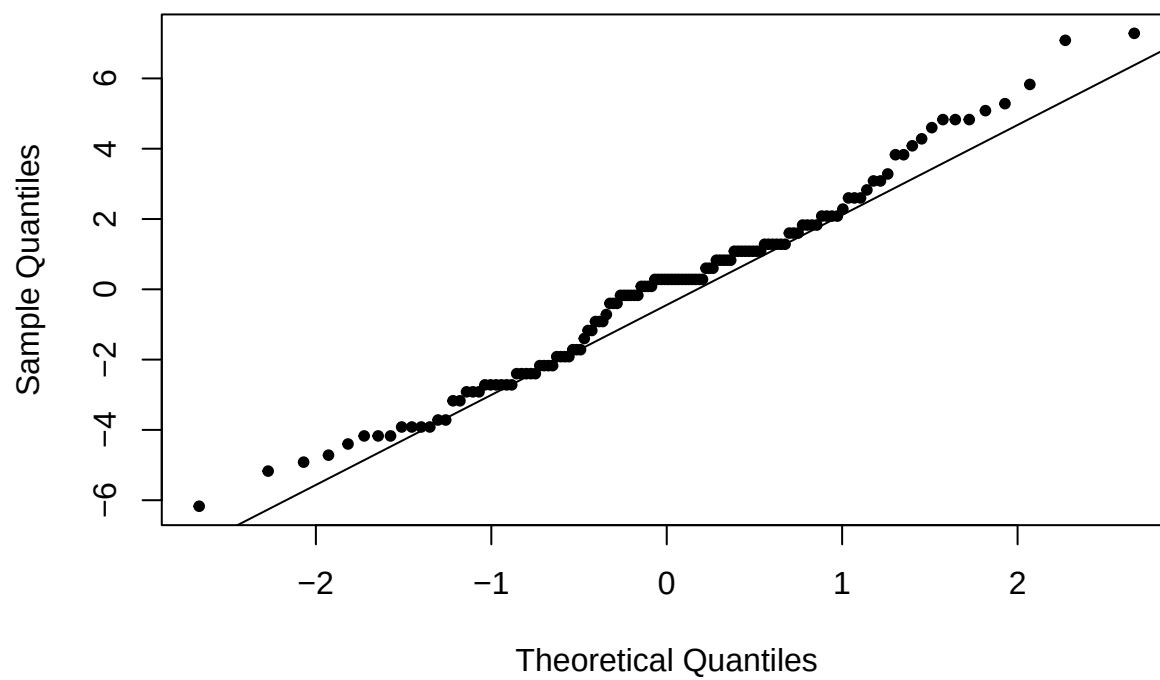
```
plot(anova_lm$fitted, anova_lm$resid,  
     xlab = "Fitted values",  
     ylab = "Residuals",  
     pch = 20)  
abline(h = 0)
```



```
#Plotting Q-Q plot
```

```
qqnorm(anova_lm$resid, pch = 20)  
qqline(anova_lm$resid)
```


Normal Q-Q Plot

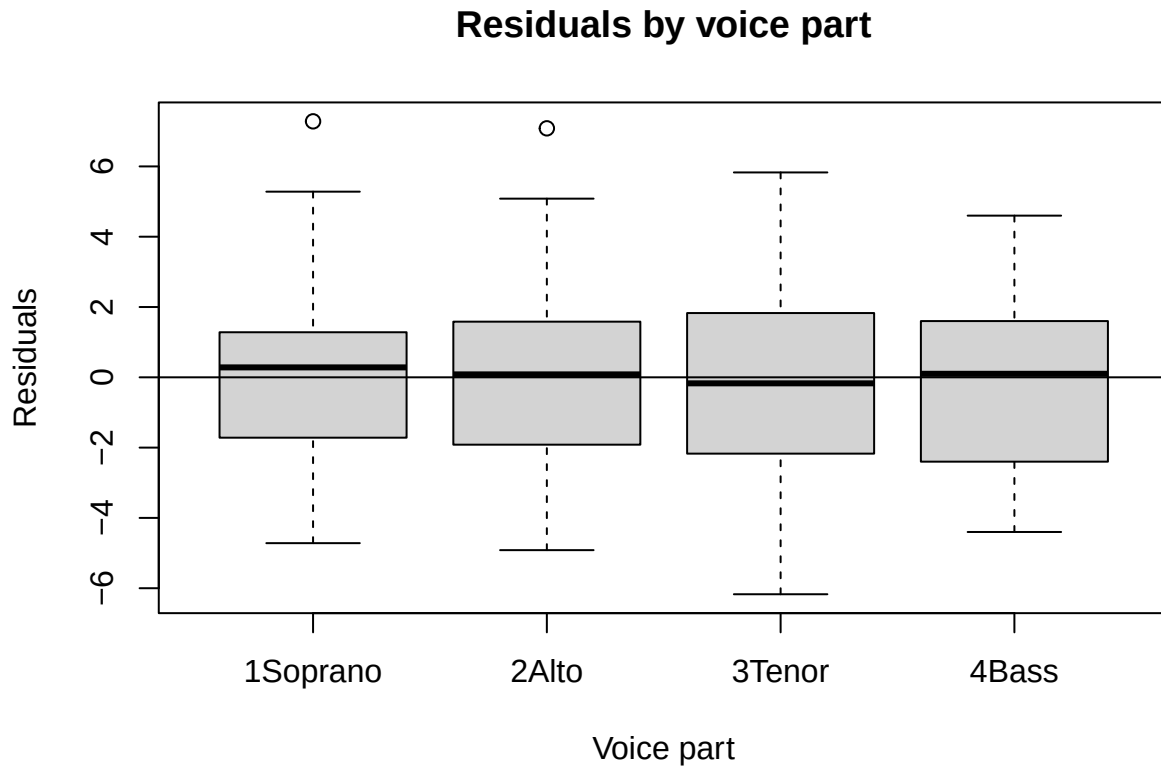


```
# Residual by voice part

singer_lm <- lm(height ~ voice, data = singers)

plot(singers$voice, singer_lm$resid,
     xlab = "Voice part",
     ylab = "Residuals",
     main = "Residuals by voice part")

abline(h = 0)
```



The residuals versus fitted values plot shows four vertical bands, corresponding to the four fitted group means. The spread of residuals appears broadly similar across the groups, so there is no strong visual evidence against the constant variance assumption. We also grouped the residuals by voice part to check whether the spread of residuals was similar within each group. The residual boxplots are all roughly centred around zero. The Soprano group has the largest deviation in median from zero, although this deviation is still small. The Tenor group appears to have a slightly wider spread than the other groups, but not to an extent that suggests serious concern. Overall, there is little evidence that the constant variance assumption is violated.

The Q-Q plot shows slight S-shaped curvature, with the largest deviations occurring in the tails. This suggests that the normality assumption is reasonably satisfied, although the tail behaviour is not ideal.

Overall, the model appears to satisfy the constant variance assumption reasonably well, while the normality assumption is acceptable but not perfect.

Additional result checks

```
test_iii <- lm(log(y) ~ x2 + x3 + x4,
               data = petrol,
               subset = 41:125)

test_p_iii_log <- predict(test_iii, newdata = petrol[1:40, ])
test_p_iii <- exp(test_p_iii_log)
t_rmse_iii <- sqrt(mean((petrol$y[1:40] - test_p_iii)^2))

test_iv <- lm(log(y) ~ x2 + x4 + I(x3^2),
               data = petrol,
               subset = 41:125)

test_p_iv_log <- predict(test_iv, newdata = petrol[1:40, ])
test_p_iv <- exp(test_p_iv_log)
t_rmse_iv <- sqrt(mean((petrol$y[1:40] - test_p_iv)^2))

print(t_rmse_iii)

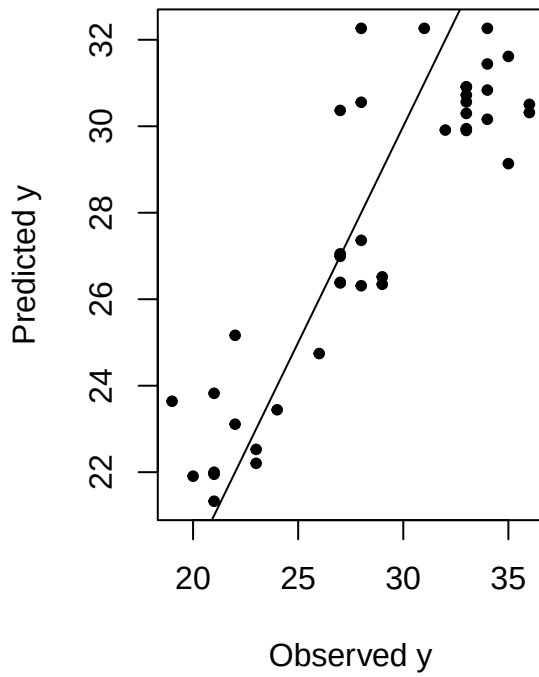
## [1] 2.725435

print(t_rmse_iv)

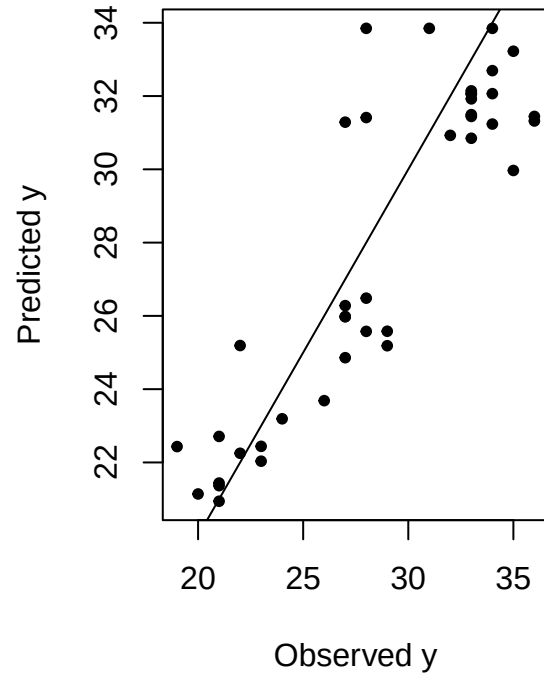
## [1] 2.480595

par(mfrow = c(1, 2))
plot(petrol$y[1:40], test_p_iii,
     xlab = "Observed y",
     ylab = "Predicted y",
     main = "Part iii model",
     pch = 20)
abline(0, 1)
plot(petrol$y[1:40], test_p_iv,
     xlab = "Observed y",
     ylab = "Predicted y",
     main = "Part iv model",
     pch = 20)
abline(0, 1)
```

Part iii model



Part iv model



```
par(mfrow = c(1, 1))
```